

Reviewer Pack — Noncommutative L^p Norm Lower Bounds for Disjointness

Entry 0acf2c6083e2 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Noncommutative L^p Norm Lower Bounds for Disjointness

Entry ID: 0acf2c6083e2 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-01 06:07:50 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative L^p Geometry **Field B** (complexity object): Randomized Communication Complexity of Disjointness

Statement:

For any $n \times n$ matrix M representing the Disjointness function, the noncommutative L^p norm $\|M\|_p \geq \Omega(\sqrt{n})$ for $p = \log n$. This bound is tight up to $\text{polylog}(n)$ factors.

Rationale (proposer’s reasoning):

Noncommutative L^p norms capture operator-space geometry of matrices, which could expose hidden structure in communication protocols. The $\log n$ choice of p aligns with known lower bounds for multi-party protocols, while avoiding standard operator norms that have been extensively studied.

Taxonomy category: LIFTING (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): f1c1a638f0a34ac1

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.95	SAFE	SAFE
ALGEBRIZATION	SAFE	0.95	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 10 # Start with a small size and increase if needed
    M = [[random.randint(0, 1) for _ in range(n)] for _ in range(n)]

    def matrix_dilation(M):
        I = [[1 if i == j else 0 for j in range(n)] for i in range(n)]
        D = []
        for row in M:
            new_row = [x * n for x in row]
            D.append(new_row)
        return D

    def noncommutative_Lp_norm(M, p):
        if p == 1:
            norm = max(sum(abs(x) for x in row) for row in M)
        elif p == float('inf'):
            norm = max(max(row) for row in M)
        else:
            norm = (sum(sum(abs(x)**p for x in row) for row in M))**(1/p)
        return norm

    D = matrix_dilation(M)
    norm_D = noncommutative_Lp_norm(D, math.log(n))

    metric_value = norm_D / math.sqrt(n)
    conjecture_holds = metric_value >= 0.1 * math.sqrt(n)
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
        "metric_name": "noncommutative_Lp_norm",
        "metric_value": metric_value,
        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 35)]
```

```

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_value = (sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))**0.5
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value:.4f} std={std_value:.4f} support_fraction={support_fraction:.4f}")
else:
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

nces_tested': 1, 'conjecture_holds': True, 'counterexample': ''
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 17.141263493487763, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 17.589390189586325, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 16.677331100759226, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 18.30487471363759, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 16.988451401307042, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 17.589390189586325, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 18.164707688084967, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 15.695162405340357, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 17.141263493487763, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 18.581092996066843, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 17.589390189586325, 'instances_tested': 1}
TRIAL: {'metric_name': 'noncommutative_Lp_norm', 'metric_value': 18.164707688084967, 'instances_tested': 1}
RESULT: SUPPORTED mean=17.1583 std=0.6962 support_fraction=1.00

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

n too small: All trials test n=1 instances, preventing observation of scaling behavior. The metric values (~17-18) don't show growth with n, contradicting the $\Omega(\sqrt{n})$ claim. With only 1 instance per trial, statistical significance is invalid.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

All trials used n=1, preventing scaling analysis. Metric values show no growth with n, contradicting $\Omega(\sqrt{n})$ claim. Statistical significance invalid with 1 instance per trial. | next: Test with varying $n \geq 1000$ to observe scaling behavior of noncommutative L^p norm

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	107553
2	propose	ollama_remote	qwen3:8b	0	0	65366
3	preregistration	ollama_remote	qwen3:8b	0	0	24182
4	novelty	ollama_remote	qwen3:8b	0	0	21023
5	novelty	ollama_remote	qwen3:8b	0	0	13296
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13121
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6950
8	critic	ollama_remote	qwen3:8b	0	0	29981
9	judge	ollama_remote	qwen3:8b	0	0	19100

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 300574 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/0acf2c6083e2.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/0acf2c6083e2.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/0acf2c6083e2.tar.gz` (if generated)